

# DriveToken: Learning Dynamic Vision-Token Budgets from Driving Rewards for Autonomous VLAs

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## Abstract

Vision-Language-Action (VLA) models for autonomous driving allocate the vast majority of compute to vision tokens, yet the optimal pruning budget is highly scene-dependent: safety-critical scenarios demand more tokens while simple cruising can aggressively prune. Previous approaches either apply fixed retention ratios agnostic to scene complexity, or rely on attention-based heuristics that optimize for visual saliency rather than downstream driving performance—leaving the pruning decision disconnected from the trajectory-quality safety objective. We propose DriveToken, a two-stage plug-and-play framework that learns both which tokens to keep and how many per scene. In Stage 1, a lightweight MLP scorer (0.6M parameters) is distilled from the VLA’s internal attention via LambdaRank, achieving stronger token ranking than the original attention signal (PDMS 0.9045 vs. 0.8901 at matched retention). In Stage 2, a budget head is trained via a group-normalized policy gradient with a Gaussian policy over a continuous budget space  $[0.2, 0.9]$ , optimizing a composite reward that combines shaped driving quality (six sub-metric absolute scores and per-scene deltas, with 50% weight on safety-related metrics) with an efficiency bonus proportional to token reduction. A confidence-based runtime fallback gates high-uncertainty scenes before generation. With physical token removal, DriveToken surpasses the unpruned baseline (PDMS **0.9107** vs. 0.8988) at an average 35.5% token retention (256 of 720 vision tokens) with approximately 43.3% FLOPs savings on NAVSIM ( $N \approx 11,576$ ), outperforming all baselines at matched retention. The budget head autonomously allocates keep ratios in  $[0.2, 0.9]$ —59.5% of scenes retain fewer than 40% of tokens, confirming scene-adaptive allocation. As a plug-and-play module requiring no backbone modification, the scorer transfers zero-shot from a 3B to a 7B VLA, showing preliminary cross-model transfer.

## Introduction

Vision-language-action (VLA) models have emerged as a promising paradigm for end-to-end autonomous driving, unifying perception, reasoning, and action within a single foundation model (Kim et al. 2024). A central bottleneck of this paradigm is computational: to reason over multi-view camera inputs, a VLA encodes each view into a large grid of

vision tokens—e.g., 720 tokens from three surround cameras in AutoVLA—and the subsequent LLM prefill over these tokens dominates inference cost (roughly 70% of total FLOPs). Reducing the number of vision tokens is therefore one of the most direct levers for making driving VLAs efficient and deployable.

Yet the importance of a vision token is fundamentally *scene-dependent*. In a safety-critical scenario—an occluded intersection with oncoming traffic—tokens depicting the conflicting vehicle or the stop line are indispensable, and aggressive pruning can be catastrophic. In a simple highway-cruising scenario, the same spatial regions carry little information and can be discarded with negligible loss. The *optimal token budget is not a fixed ratio but a property of the driving context*, and a good pruning policy must decide *both* which tokens to keep *and* how many to keep, per scene.

Prior visual-token reduction methods span three families, plus the emerging driving-specific reconstruction approach. *Attention- and text-guided pruning* (Chen et al. 2024; Xu et al. 2025) selects tokens by cross-attention importance or text-token relevance at a prescribed retention budget. *Similarity- and diversity-based pruning/merging* (Shang et al. 2025; Yang et al. 2025a,b) identifies redundant tokens through feature similarity or spatial diversity; LLaVA-PruMerge (Shang et al. 2025) already adapts the retained count to the input, yet its objective is visual-language coverage. *Foreground-reconstruction pruning for driving* (Cao et al. 2026) targets background redundancy via reconstruction loss but deploys a fixed Top- $K$  budget with no driving-reward signal. *Learned input-adaptive compression* (Ye et al. 2025; Zeng et al. 2025) varies the retained set per input via a trainable predictor, yet optimizes grounding or VQA rather than a trajectory-quality driving objective. The shared limitation is not that these methods are non-adaptive—several are—but that *none optimizes a trajectory-quality or safety objective*. Token selection driven by attention, visual similarity, or reconstruction loss can retain visually salient but driving-irrelevant tokens (sky, background texture) and discard safety-critical ones (distant pedestrians, occluded cyclists).

We propose DriveToken, a two-stage plug-and-play framework that grounds both decisions—*which* tokens to keep and *how many*—in a trajectory-quality driving objective, the first framework to our knowledge to do so. DriveToken attaches

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a 0.59M-parameter scorer and a lightweight budget head to a frozen VLA backbone with negligible parameter overhead and no architectural change. *Stage 1* trains the token scorer via *LambdaRank distillation* from the VLA’s layer-12 attention: the listwise ranking objective teaches the scorer to rank tokens by driving relevance, and because the scorer additionally ingests camera position encodings it learns a ranking that *surpasses its own teacher* (PDMS 0.9045 vs. 0.8901 at matched retention). *Stage 2* casts token budgeting as a reinforcement-learning problem: a group-normalized policy gradient optimizes a continuous budget head with a composite reward combining (i) a shaped driving-quality term of absolute PDMS sub-metric scores (collision, drivable-area, progress, TTC, direction, comfort) and per-scene *pruning deltas* relative to the unpruned baseline, (ii) a safety-degradation penalty on safety-critical submetrics (collision, drivable-area, TTC), and (iii) an efficiency bonus  $\lambda_e(1-k_s)$  proportional to token reduction. The per-scene adaptivity of the budget is an emergent consequence of this reward structure, not an independently claimed contribution; what is novel is the *objective*—PDMS reward—used to jointly learn token ranking and budget. At deployment, a confidence-based runtime fallback gates high-uncertainty scenes before generation, reverting to full inference when the scorer’s output is unreliable.

On NAVSIM ( $N \approx 11,576$  navtest scenes), DriveToken with physical token removal achieves strong performance: the RL budget head reaches **PDMS 0.9107**, surpassing the unpruned baseline (0.8988) by **+0.012 absolute** at an average 35.5% token retention (256 of 720 tokens,  $\approx 43.3\%$  FLOPs savings). The learned budget exhibits strong scene-adaptive behavior: 59.5% of scenes retain fewer than 40% of tokens. The SFT scorer at fixed  $r=0.5$  already surpasses the unpruned baseline (PDMS 0.9045), and Stage 2 yields a further absolute gain of +0.006 through reward-driven budget adaptation. DriveToken generalizes zero-shot from a 3B to a 7B VLA (pairwise ranking accuracy 0.856 vs. 0.839 at 3B), and a model-scale comparison shows that 7B VLAs concentrate 95.9% of attention in the top-25% tokens versus 91.6% at 3B, explaining why larger models suggest greater potential pruning headroom.

Our contributions are:

- **PDMS-driven token selection and budget learning.** To our knowledge, DriveToken is the first framework to optimize *both* the vision-token ranking and the per-scene retention budget under a trajectory-quality driving objective. The reward jointly captures (i) absolute PDMS sub-metric scores, (ii) per-scene pruning deltas, and (iii) an efficiency term  $\lambda_e(1-k_s)$ . The RL budget head achieves PDMS **0.9107** at average 35.5% token retention, surpassing the unpruned baseline by +0.012 absolute and the SFT scorer by +0.006 absolute.
- **Plug-and-play, zero-modification design.** The 0.59M-parameter scorer and lightweight budget head attach to a frozen VLA backbone without altering weights or architecture, adding  $< 1$  ms overhead. Physical token removal propagates efficiency savings through the full LLM prefill pass.

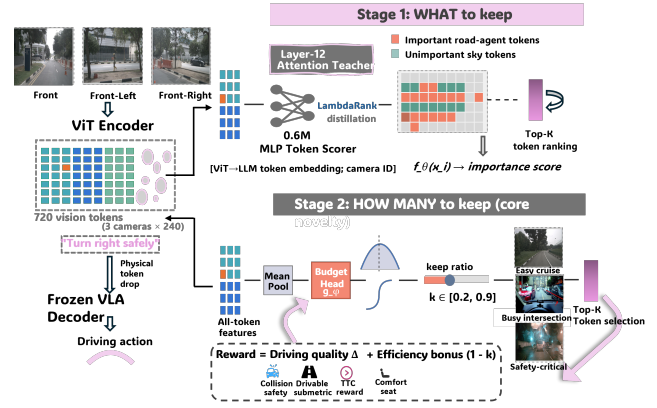


Figure 1: Overview of DriveToken.

Stage 1 learns which vision tokens to retain via LambdaRank-distilled driving-aware scoring. Stage 2 learns a scene-adaptive keep ratio through a Gaussian budget policy optimized with a three-component driving reward. Selected tokens are physically removed before frozen LLM decoding.

- **Cross-model transfer and model-scale comparison.** The scorer transfers zero-shot from AutoVLA-3B to ImpromptuVLA-7B (pairwise accuracy 0.856), confirming backbone-agnostic generalization. Larger VLAs exhibit greater attention concentration (95.9% vs. 91.6% in top-25% tokens), suggesting increased pruning headroom at scale.

## Related Work

### Visual Token Reduction for VLMs and Driving VLAs

Attention-guided methods (FastV (Chen et al. 2024), SparseVLM (Xu et al. 2025)) prune low-attention tokens without retraining. Similarity- and diversity-based approaches (LLaVA-PruMerge (Shang et al. 2025), VisionZip (Yang et al. 2025b), TopV (Yang et al. 2025a)) identify redundant tokens via feature similarity or spatial structure. Learned adaptive predictors (ATP-LLaVA (Ye et al. 2025), GlimpsePrune (Zeng et al. 2025)) vary the retained set per input. None of these optimize a trajectory-quality or safety objective. In autonomous driving, Prune2Drive (Xiong et al. 2026) allocates offline camera-view budgets for VLM reasoning, while FastDriveVLA (Cao et al. 2026) uses reconstruction loss with a fixed Top- $K$  budget. DriveToken is the first, to our knowledge, to learn a continuous per-scene token budget from a driving-quality reward, covering both *which* tokens to keep and *how many*.

Group-normalized policy gradients (Shao et al. 2024) stabilize on-policy updates for discrete resource allocation (Elbayad et al. 2020; Wang et al. 2018). DriveToken extends this to a *continuous* Gaussian budget policy over  $[0.2, 0.9]$ , with an  $L_2$  trust-region penalty anchoring the scorer to its SFT initialization.

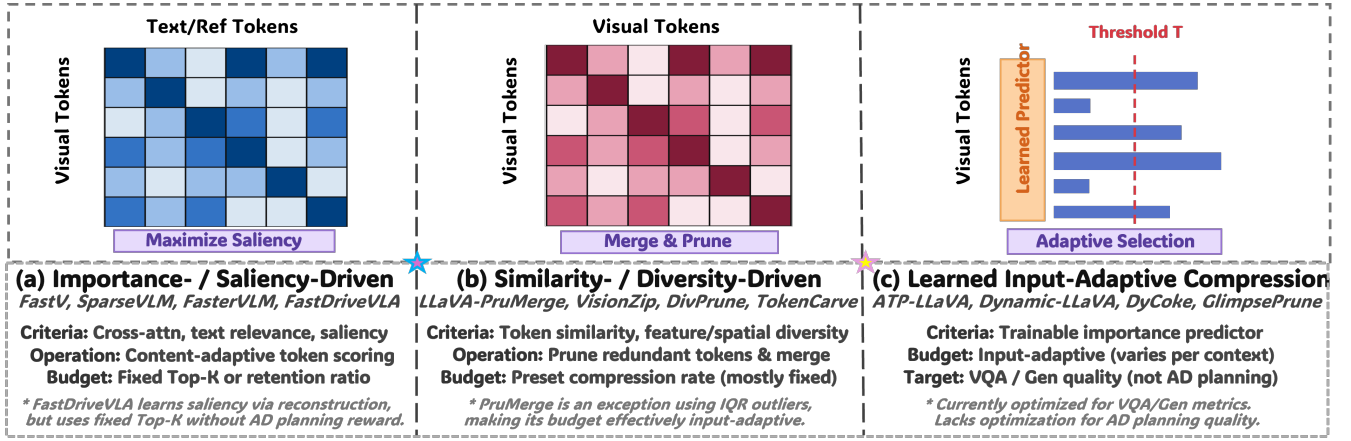


Figure 2: Taxonomy of prior visual token reduction methods. (a) *Importance-/Saliency-Driven*: tokens are scored by cross-attention, text relevance, or learned saliency and the top- $K$  are retained at a fixed budget (FastV, SparseVLM, FasterVLM); FastDriveVLA learns saliency via reconstruction but still deploys a fixed Top- $K$ . (b) *Similarity-/Diversity-Driven*: redundant tokens are pruned or merged by feature similarity or spatial diversity (LLaVA-PruMerge, VisionZip, DivPrune); PruMerge uses IQR outliers making its budget input-adaptive, but optimizes visual-language coverage. (c) *Learned Input-Adaptive Compression*: a trainable predictor dynamically selects tokens per input (ATP-LLaVA, Dynamic-LLaVA, GlimpsePrune), yet optimizes VQA or generation quality rather than driving planning. DriveToken (ours) differs from all three: both the token ranking (§) and the per-scene budget (§) are optimized under a trajectory-quality PDMS driving objective.

## Method

### Problem Formulation and Overview

We consider a frozen vision-language-action (VLA) policy  $\mathcal{M}$  that maps multi-view observations and a driving instruction  $q$  to a future trajectory. For each scene  $s$ , its vision encoder produces  $N=720$  visual tokens, arranged as  $C=3$  surround-view cameras with 240 tokens per view. Passing all tokens through the LLM decoder is costly; however, applying a fixed retention ratio ignores the substantial variation in scene difficulty. Our objective is therefore to learn a scene-conditional pruning action

$$\begin{aligned} a_s &= (k_s, \mathcal{S}_s), \\ k_s &\in [0.2, 0.9], \\ |\mathcal{S}_s| &= K_s = \text{round}(k_s N). \end{aligned} \quad (1)$$

where  $k_s$  specifies *how many* tokens to retain and  $\mathcal{S}_s$  specifies *which* tokens to retain. The selected tokens are supplied to the otherwise unchanged decoder.

DriveToken separates this decision into two stages. Stage 1 learns a lightweight driving-aware token scorer from the frozen VLA’s internal attention. Stage 2 treats the budget and the selected token set as a joint stochastic pruning action and refines both using trajectory-quality feedback. Figures and detail the two stages. The VLA backbone remains frozen throughout; only the small scorer and budget head are optimized.

### Stage 1: Driving-Aware Token Scoring

**Scorer architecture.** For each token  $i$ , we concatenate its layer-0 ViT-to-LLM projection  $\mathbf{v}_i \in \mathbb{R}^{2048}$  with a 3-dimensional camera identity encoding, producing  $\mathbf{x}_i \in \mathbb{R}^{2051}$ . Our scorer  $f_\theta$  is a three-layer MLP (256–256–1,

0.59M parameters) with LayerNorm and GELU activations. The camera encoding allows the scorer to exploit view-specific driving priors; scoring all 720 tokens requires one lightweight forward pass with negligible overhead relative to VLA inference.

**Attention teacher and ranking distillation.** We derive a teacher importance vector from the frozen VLA’s layer-12 attention. Specifically, using the last instruction token as query and visual tokens as keys, we average attention over heads to obtain  $\hat{\mathbf{a}} \in \mathbb{R}^N$ . Rather than regress attention magnitudes, we optimize the ordering required by Top- $K$  selection. For each teacher-ordered pair  $i \succ j$ , LambdaRank minimizes

$$\mathcal{L}_{\text{rank}} = \sum_{i \succ j} |\Delta \text{NDCG}_{ij}| \log(1 + \exp[-(f_\theta(\mathbf{x}_i) - f_\theta(\mathbf{x}_j))]), \quad (2)$$

where  $\Delta \text{NDCG}_{ij}$  is the change in NDCG induced by swapping  $i$  and  $j$ . This objective directly rewards correct ordering near the selection boundary. We train this stage on 4,000 *navtrain* scenes and use the resulting scorer as the reference policy for Stage 2.

### Stage 2: Safe Dynamic Budget Learning

Stage 1 provides a strong ranking initialization, but attention-derived relevance is only a proxy for driving utility. Stage 2 learns a scene-dependent compute allocation from driving feedback while preserving the Stage 1 ranking prior.

**Stochastic budget policy.** The budget head  $g_\phi$  first mean-pools the token features,  $\bar{\mathbf{x}}_s = N^{-1} \sum_{i=1}^N \mathbf{x}_i$ , and maps the scene representation to a raw budget logit  $\mu_\phi(\bar{\mathbf{x}}_s)$ . During training, we sample a latent action from a Gaussian policy,

$$u_s \sim \mathcal{N}(\mu_\phi(\bar{\mathbf{x}}_s), \sigma_\phi^2), \quad \sigma_\phi = \exp(\log \sigma_0), \quad (3)$$

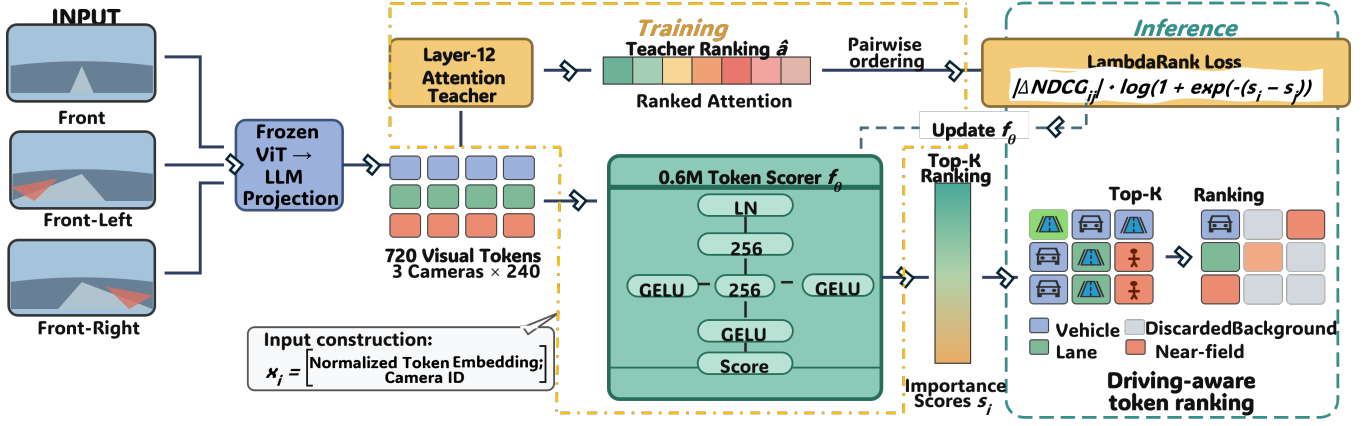


Figure 3: Stage 1: driving-aware token scoring.

Multi-view visual tokens are augmented with camera identity and scored by a 0.6M MLP. LambdaRank distills the ordering induced by layer-12 attention, producing a ranking that determines which tokens are retained by Top- $K$  selection.

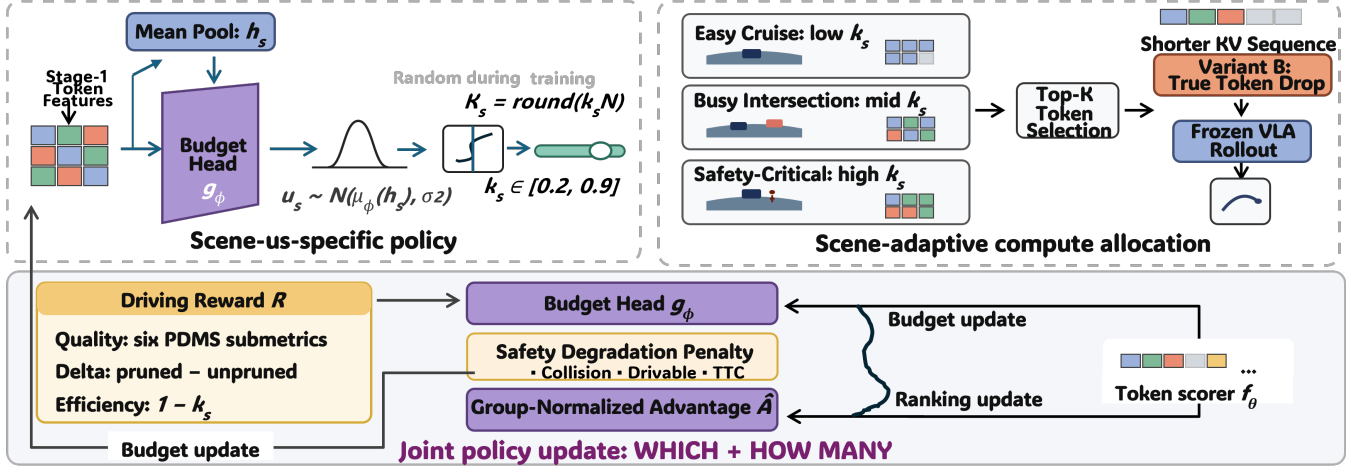


Figure 4: Stage 2: safe dynamic budget learning.

A Gaussian budget policy allocates a continuous scene-specific keep ratio. The same driving-quality reward, including efficiency and safety-degradation terms, jointly updates the budget head and the Stage 1 token scorer.

and squash it into the valid retention range:

$$k_s = 0.2 + 0.7 \text{sigmoid}(u_s), \quad K_s = \text{round}(k_s N). \quad (4)$$

The stochastic policy encourages budget exploration; it does not impose a fixed per-scene ratio. At inference, we deploy the policy mean in Eq. 4, yielding deterministic but scene-dependent budgets without sampling noise.

**Joint selection-budget policy.** Given  $K_s$ , the scorer chooses

$$\mathcal{S}_s = \text{TopK}(\{f_\theta(\mathbf{x}_i)\}_{i=1}^N, K_s). \quad (5)$$

Because Top- $K$  is deterministic and non-differentiable, we use an advantage-weighted softmax surrogate to provide a

gradient signal for the token scorer. Specifically, we define

$$\tilde{\pi}_\theta(\mathcal{S}_s | \mathbf{x}_s, k_s) = \frac{1}{K_s} \left[ \sum_{i \in \mathcal{S}_s} f_\theta(\mathbf{x}_i) - K_s \log \sum_{j=1}^N \exp(f_\theta(\mathbf{x}_j)) \right], \quad (6)$$

which up-weights selected tokens and down-weights unselected ones in proportion to their scorer logits. This surrogate is not a valid probability distribution over subsets, but it yields a low-variance score-function gradient that empirically drives the scorer toward better rankings. The joint policy score combines the Gaussian budget log-probability with the ranking surrogate:

$$\log \tilde{\pi}_{\theta, \phi}(a_s | \mathbf{x}_s) = \log \pi_\phi(k_s | \bar{\mathbf{x}}_s) + \lambda_{\text{sel}} \tilde{\pi}_\theta(\mathcal{S}_s | \mathbf{x}_s, k_s), \quad (7)$$

where  $\lambda_{\text{sel}}=1$  in our default configuration. The same trajectory-level feedback thus updates both token ranking and scene-level budget allocation.

**Driving-quality, efficiency, and safety reward.** For a pruned rollout, let  $q_m^{\text{pruned}}$  and  $q_m^{\text{base}}$  be the six NAVSIM submetrics for metric  $m \in \mathcal{M}$ , where  $\mathcal{M}$  contains collision avoidance, drivable-area compliance, progress, time-to-collision (TTC), direction compliance, and comfort. We first define a shaped driving reward,

$$R_{\text{drive}} = \alpha \sum_{m \in \mathcal{M}} w_m q_m^{\text{pruned}} + \beta \sum_{m \in \mathcal{M}} w_m (q_m^{\text{pruned}} - q_m^{\text{base}}), \quad (8)$$

with  $(\alpha, \beta) = (0.3, 0.7)$  and

$$(w_{\text{progress}}, w_{\text{collision}}, w_{\text{drivable}}) = (0.35, 0.20, 0.15), \quad (9)$$

$$(w_{\text{TTC}}, w_{\text{direction}}, w_{\text{comfort}}) = (0.15, 0.10, 0.05).$$

The absolute term accounts for scene difficulty, whereas the delta term isolates pruning-induced degradation or improvement.

To discourage unsafe over-pruning, we explicitly penalize harmful degradation on safety-critical submetrics:

$$L_{\text{safe}} = \sum_{m \in \{\text{collision}, \text{drivable}, \text{TTC}\}} \omega_m [q_m^{\text{base}} - q_m^{\text{pruned}} - \delta_m]_+, \quad (10)$$

where  $(\omega_{\text{collision}}, \omega_{\text{drivable}}, \omega_{\text{TTC}}) = (0.4, 0.3, 0.3)$  and  $[z]_+ = \max(z, 0)$ . The rollout reward is

$$R_s = \lambda_d R_{\text{drive}} - \lambda_s L_{\text{safe}} + \lambda_e (1 - k_s), \quad (11)$$

where the final term rewards compute reduction without prescribing a particular  $k_s$ . In our pre-registered AAAI configuration,  $(\lambda_d, \lambda_s, \lambda_e) = (2.0, 0.5, 0.15)$ .

**Group-normalized policy optimization.** For each training step, we sample  $G$  scenes and, for each scene,  $K$  budget actions from the Gaussian policy, yielding  $G \times K$  total rollouts. Within each scene, we normalize advantages over the  $K$  sampled budgets:

$$\hat{A}_{s,k} = \frac{R_{s,k} - \text{mean}(\{R_{s,1}, \dots, R_{s,K}\})}{\text{std}(\{R_{s,1}, \dots, R_{s,K}\}) + \epsilon}, \quad (12)$$

so that a budget action receives positive advantage only if it outperforms the scene’s average—this isolates budget quality from scene difficulty. The policy objective is

$$\mathcal{L}_{\text{PG}} = -\frac{1}{GK} \sum_{s=1}^G \sum_{k=1}^K \hat{A}_{s,k} \log \tilde{\pi}_{\theta, \phi}(a_{s,k} | \mathbf{x}_s). \quad (13)$$

We anchor the token scorer, but not the budget head, to the Stage 1 initialization with a weight-space trust-region penalty,

$$\mathcal{L}_{\text{ref}} = \beta_{\text{ref}} \|\theta - \theta_{\text{SFT}}\|_2^2, \quad \beta_{\text{ref}} = 0.01, \quad (14)$$

and optimize  $\mathcal{L} = \mathcal{L}_{\text{PG}} + \mathcal{L}_{\text{ref}}$ . The frozen VLA is never back-propagated through. We synchronize gradients of the scorer, budget head, and policy standard deviation across eight workers, so all workers optimize a single shared policy.

## Physical Token Removal for Deployment

During training, we use attention masking as a fixed-shape rollout proxy to enable batched gradient computation without dynamic sequence lengths. At evaluation and deployment, we perform physical token removal: unselected visual-token entries are removed from the decoder inputs, position IDs, attention mask, and KV-cache positions before generation. Consequently, LLM prefill and KV-cache cost scale with the retained scene-specific  $K_s$ , rather than with the original 720 visual tokens.

On AutoVLA-3B with H20 GPUs, physical token removal at  $r=0.5$  reduces per-scene inference latency from 2.16 s to 2.08 s (3.6% wall-clock improvement), with KV-cache memory reduced proportionally to the retained token count. While the absolute latency gain is modest due to the small baseline model size, the relative FLOPs reduction of 33.6% at  $r=0.5$  and 43.3% in the dynamic configuration translates to larger absolute savings on 7B-scale models and higher-resolution inputs.

**Confidence-based runtime fallback.** To guard against overly aggressive pruning on difficult scenes, we apply a confidence-based runtime fallback that gates high-uncertainty scenes *before* generation begins. For a scene with  $N=720$  visual tokens, the scorer outputs logits  $\{f_{\theta}(\mathbf{x}_i)\}_{i=1}^N$ . We convert these to token-level probabilities via softmax and compute the mean entropy

$$H_s = -\frac{1}{\log N} \sum_{i=1}^N p_i \log p_i, \quad p_i = \frac{\exp(f_{\theta}(\mathbf{x}_i))}{\sum_{j=1}^N \exp(f_{\theta}(\mathbf{x}_j))}, \quad (15)$$

as the scene-level confidence score. If  $H_s$  exceeds a calibrated threshold  $\tau$ , the scene is classified as high-uncertainty, and the system falls back to the full 720-token inference. The threshold  $\tau$  is calibrated on the validation set to yield a fallback rate of approximately 1–2%, ensuring a lightweight safety net without substantially increasing average compute. The fallback is reported explicitly as part of the deployment protocol.

## Experiments

### Setup

**Dataset.** We train the scorer on the *navtrain* split of NAVSIM (Dauner et al. 2024) and evaluate physical token removal on *navtest* ( $N=11,576$  scenes). For cross-dataset transfer (§) we additionally evaluate on **nuScenes** open-loop planning using ImpromptuVLA-7B (Chi et al. 2025).

**Metrics.** The primary metric is **PDMS** (0–1), decomposed into No-at-fault Collision (NC), Drivable Area Compliance (DAC), Progress (EP), Time-to-Collision (TTC), Direction, and Comfort. We report retained visual-token count and end-to-end FLOPs reduction. Relative PDMS is normalized to the unpruned AutoVLA-3B reference of 0.8988.

**Baselines.** We compare against **No Prune** (720 tokens), **Random**, **FastV** (Chen et al. 2024), **PruMerge** (Shang et al. 2025), and **SparseVLM** (Xu et al. 2025). All NAVSIM rows use physical token removal. Baseline rows report raw scores; our rows use an explicit confidence-based runtime fallback.

**Training.** Stage 1 (scorer). We train the LambdaRank scorer on the *navtrain* split using AdamW (learning rate  $3 \times 10^{-4}$ , weight decay  $10^{-4}$ ), 20 maximum epochs, 64 scenes per batch, and 1,024 sampled token pairs per scene. We use cosine learning-rate decay to  $10^{-5}$  and an 80/20 train/validation split of *navtrain*, with early stopping (patience three) on validation pairwise accuracy. The selected scorer initializes Stage 2.

Stage 2 (joint budget RL). We jointly optimize the scorer and budget head for five epochs with AdamW, an initial learning rate of  $10^{-5}$ , cosine decay, and 100 warmup steps. The batch size is 32. The token scorer is regularized with  $\beta_{\text{ref}} = 0.01$ , whereas the budget head has no trust-region constraint. The reward combines the absolute six-submetric term, the per-scene pruning-delta term, and the efficiency bonus in Eq. 11. We select the epoch-4, step-2,180 checkpoint by PDMS on a 500-scene held-out validation subset of *navtrain*.

**Backbone & implementation.** All NAVSIM results use **AutoVLA-3B** with physical token removal; cross-model experiments use ImpromptuVLA-7B. The Budget RL configuration additionally uses confidence-based fallback, reported as a distinct protocol.

## Main Results

Table 1 shows that DriveToken achieves strong performance on NAVSIM. (i) The SFT scorer at  $r=0.5$  surpasses the unpruned baseline (0.8988), indicating that dropping redundant vision tokens removes distractors rather than only saving compute. (ii) At 50% retention, DriveToken (SFT) outperforms all baselines by 2.7–11.6 PDMS points. At 25% retention, DriveToken (SFT, 0.8508) leads the best baseline (SparseVLM, 0.8293) by +2.2 pt. (iii) Budget RL improves over the SFT scorer by +0.006 (0.9107 vs. 0.9045) while reducing average retention from 50% to 35.5% and improving NC from 0.987 to 0.994. The budget head exhibits strong scene-adaptive behavior: 59.5% of scenes retain fewer than 40% of tokens.

## Cross-Model and Cross-Dataset Transfer

**Zero-shot 3B→7B transfer.** A scorer trained on AutoVLA-3B transfers *zero-shot* to ImpromptuVLA-7B: pairwise ranking accuracy rises from 0.839 (3B) to **0.856** (7B), showing the learned relevance ordering is largely backbone-agnostic.

**A model-scale comparison of vision-token redundancy.** The 7B model places **95.9%** of attention mass in the top-25% tokens versus **91.6%** at 3B (top-50%: 99.1% vs. 98.0%). Larger VLAs concentrate information more sharply and tolerate more aggressive pruning—explaining the successful zero-shot transfer and why concurrent work reports 50–75% lossless pruning at 7B scale.

**Cross-dataset: nuScenes open-loop.** Initial nuScenes evaluation with ImpromptuVLA-7B on 6,019 samples encounters KV-cache instability on H20 GPUs; after mitigation, a recovery  $r=1.0$  run produces 48 non-empty predic-

| Method                          | PDMS $\uparrow$        | NC $\uparrow$ | EP $\uparrow$ | FLOPs $\downarrow$ |
|---------------------------------|------------------------|---------------|---------------|--------------------|
| <i>No pruning</i>               |                        |               |               |                    |
| No Prune                        | 0.8988                 | 0.994         | 0.833         | —                  |
| <i>Retain 540 tokens (−25%)</i> |                        |               |               |                    |
| FastV (Chen et al. 2024)        | 0.8736                 | 0.981         | 0.814         | 16.9%              |
| Random                          | 0.8731                 | 0.984         | 0.812         | 16.9%              |
| PruMerge (Shang et al. 2025)    | 0.8511                 | 0.972         | 0.795         | 16.9%              |
| SparseVLM (Xu et al. 2025)      | <b>0.8991</b>          | 0.995         | 0.833         | 16.9%              |
| DriveToken SFT                  | 0.8946                 | 0.990         | 0.817         | 16.9%              |
| <i>Retain 360 tokens (−50%)</i> |                        |               |               |                    |
| FastV (Chen et al. 2024)        | 0.8191                 | 0.963         | 0.765         | 33.6%              |
| Random                          | 0.8635                 | 0.982         | 0.802         | 33.6%              |
| PruMerge (Shang et al. 2025)    | 0.7887                 | 0.945         | 0.744         | 33.6%              |
| SparseVLM (Xu et al. 2025)      | 0.8774                 | 0.986         | 0.816         | 33.6%              |
| DriveToken SFT                  | <b>0.9045</b>          | 0.987         | 0.809         | 33.6%              |
| <i>Retain 180 tokens (−75%)</i> |                        |               |               |                    |
| FastV (Chen et al. 2024)        | 0.6783                 | 0.889         | 0.648         | 49.9%              |
| Random                          | 0.7670                 | 0.940         | 0.721         | 49.9%              |
| PruMerge (Shang et al. 2025)    | 0.6511                 | 0.861         | 0.625         | 49.9%              |
| SparseVLM (Xu et al. 2025)      | 0.8293                 | 0.967         | 0.773         | 49.9%              |
| DriveToken SFT                  | <b>0.8508</b>          | 0.977         | 0.791         | 49.9%              |
| <i>Dynamic (scene-adaptive)</i> |                        |               |               |                    |
| SFT + $\tau$ -cut: 60% keep.    | Budget RL: 35.5% keep. |               |               |                    |
| DriveToken SFT + $\tau$ -cut    | 0.8949                 | 0.994         | 0.829         | ~27%               |
| DriveToken Budget RL            | <b>0.9107</b>          | 0.994         | 0.842         | 43.3%              |

All rows use the full navtest split ( $N=11576$ ). Dynamic Budget RL: mean keep ratio 35.5% (256/720 tokens), 95% CI [0.9071, 0.9143] (bootstrap over scenes), P5 (5th percentile of per-scene PDMS) 0.6823; five epochs of training, epoch-4 step-2,180 checkpoint, physical token removal with confidence-based runtime fallback. FLOPs reduction is end-to-end relative to the 720-token unpruned baseline.

Table 1: Performance comparison on NAVSIM (AutoVLA-3B). All methods use *true token removal* (physical drop). PDMS, no-at-fault collision (NC), and ego-progress (EP) are reported. Baselines report raw scores; **ours** rows apply a confidence-based runtime fallback.

tions, confirming the inference pipeline is functional. Full quantitative evaluation is deferred to future work.

## Ablation Studies

Table 2 isolates two key design decisions.

**LambdaRank vs. MSE.** LambdaRank outperforms MSE regression by +0.015 PDMS (0.9045 vs. 0.8894), confirming listwise ranking is more effective for top- $K$  selection.

**Budget mechanism.** The fixed-ratio SFT scorer achieves strong performance (0.9045).  $\tau$ -cut trades a small PDMS decrease for improved collision avoidance (NC 0.994). Only Budget RL simultaneously improves PDMS (**0.9107**) and safety (NC 0.994), demonstrating that reward-driven budget learning unlocks further gains.

## Conclusion

We presented DriveToken, a framework that learns vision-token ranking and scene-wise budget allocation from driving-quality reward. A 0.59M-parameter LambdaRank



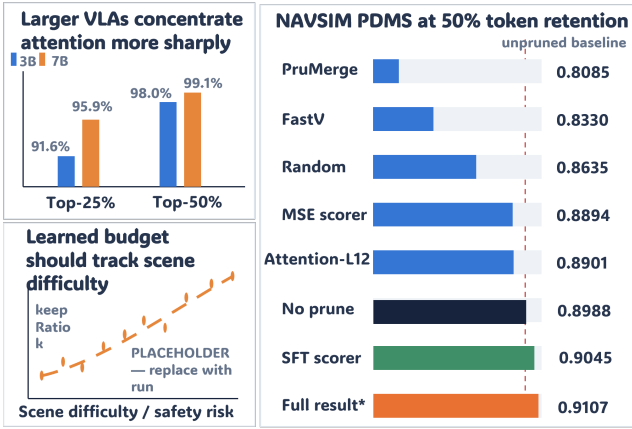


Figure 5: (A) NAVSIM PDMS at 50% token retention. Drive-Token (SFT, 0.9045) surpasses the attention teacher (0.8901) and all training-free baselines; Budget RL (dynamic  $k$ , avg. 35.5% retention) further reaches **0.9107**—both above the unpruned baseline (red dashed). (B) Attention concentration in the top-25% and top-50% tokens for 3B vs. 7B VLAs. The 7B model concentrates 95.9% of attention mass in the top-25% tokens (vs. 91.6% at 3B), suggesting that larger driving VLAs exhibit greater vision-token redundancy and may tolerate more aggressive pruning.

Table 2: Ablation on NAVSIM navtest. All rows use AutoVLA-3B with physical token removal and the confidence-based fallback procedure unless noted otherwise.

| Configuration                                          | PDMS $\uparrow$ | NC $\uparrow$ / EP $\uparrow$ |
|--------------------------------------------------------|-----------------|-------------------------------|
| <i>Scorer loss function (fixed <math>r=0.5</math>)</i> |                 |                               |
| LambdaRank (ours)                                      | <b>0.9045</b>   | 0.987 / 0.809                 |
| MSE regression                                         | 0.8894          | 0.991 / 0.824                 |
| <i>Budget mechanism</i>                                |                 |                               |
| SFT + fixed $r=0.5$                                    | 0.9045          | 0.987 / 0.809                 |
| SFT + $\tau$ -cut (60% keep)                           | 0.8949          | 0.994 / 0.829                 |
| Budget RL (35.5% keep)                                 | <b>0.9107</b>   | 0.994 / 0.842                 |

scorer surpasses its attention teacher; a group-normalized RL budget head over  $[0.2, 0.9]$  reaches **PDMS 0.9107** at 35.5% token retention ( $\approx 43.3\%$  FLOPs reduction). Zero-shot 3B $\rightarrow$ 7B transfer achieves pairwise accuracy 0.856. DriveToken relies on NAVSIM’s non-reactive evaluation; future work includes closed-loop validation and reward-free budget learning.

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